WIDE VIEW AND LINE FILTERS

Parameter Optimization, Benchmarking, and Noise Robustness

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OUTLINE

1. Introduction
2. Implementation
3. Methods
4. Parameter Ablation
5. Benchmarking
6. Noise Robustness
7. Statistical Validation
8. Discussion
9. Conclusion

INTRODUCTION

The edge detection problem and motivation

EDGE DETECTION TODAY

Edge detection has evolved through three generations of methods.

1. Classical operators (Sobel, Prewitt, Canny): fixed-size kernels, simple but limited adaptability
2. Training-free extended filters (WVF, LF): local polynomial fitting over large neighborhoods, no learned parameters
3. Deep learning (DexiNed, TEED, PiDiNet, NBED, DiffusionEdge): learned edge representations, highest clean-data scores

Bagan and Wang's Wide View Filter (WVF) and Line Filter (LF) sit between classical simplicity and DL complexity — but are their published parameters optimal?

RESEARCH QUESTIONS

This study addresses four questions through a comprehensive four-part investigation.

1. Are the published WVF/LF parameters ($N_p = 250$, $N_s = 18$, $d = 4$) optimal for standard benchmarks?
2. How do optimally-tuned WVF/LF compare against 54 classical operators and 5 DL models?
3. Does the WVF's extended spatial support provide noise robustness advantages over learned models?
4. Are these findings statistically validated across datasets?

Total scope: 500,000+ individual filter evaluations across 330 images and 4 datasets.

IMPLEMENTATION

Ground-up reimplementation with GPU acceleration

REIMPLEMENTATION

Both the WVF and LF were reimplemented from scratch based on the original papers by Bagan and Wang. No source code was available — the entire codebase was built from the mathematical formulations alone.

The implementation includes:

- CPU reference implementations for per-pixel WVF and LF (correctness validation)
- GPU-accelerated backends via PyTorch CUDA for both WVF and LF
- Shared mathematical core: Taylor design matrix construction, circular neighbor selection, coordinate rotation, pseudoinverse computation
- 11 classical baseline families (54 configurations) also GPU-accelerated via batched `conv2d`
- Full ODS/OIS evaluation pipeline with vectorized threshold sweeps

GPU ACCELERATION STRATEGY

The key insight: the per-pixel polynomial fit is embarrassingly parallel.

1. Precompute pseudoinverses: for each orientation θ_k, compute $P_\theta = (A^\top A)^{-1} A^\top$ once. Stack all N_s pseudoinverses into a single tensor of shape (N_s, M, N_p).
2. Vectorized gather: collect all N_p neighbor intensities for every interior pixel simultaneously into a (N_{pixels}, N_p) tensor F.
3. Batched matmul: compute the normal derivative at all pixels in one operation per orientation: response $= |P_\theta[1, :] \cdot F^\top|$.
4. Orientation max: take the element-wise maximum across orientations.

Memory-safe batching splits pixels into configurable chunks for large images or limited VRAM.

CPU VS. GPU PERFORMANCE

Benchmarked CPU rate: 1,365 μs/pixel ($N_p = 15$, $N_s = 18$, $d = 4$, 32×32 crop).

Image	CPU (est.)	GPU	Speedup
BSDS500 (481×321)	3.5 min	0.009 s	$23{,}000\times$
BIPED (1280×720)	21 min	0.036 s	$35{,}000\times$

- CPU: triple-nested loop — each pixel solves N_s independent `np.linalg.lstsq` calls
- GPU: precomputed pseudoinverses + single batched matmul per orientation
- GPU–CPU numerical agreement: relative error $< 10^{-6}$
- Full ablation (453,420 configs): 12.4 hours on GPU vs. $\sim$18 years on CPU

METHODS

WVF and LF formulations

WIDE VIEW FILTER (WVF)

For a target pixel $\mathbf{p}$, the WVF selects N_p nearest neighbors within a circular support. At each orientation θ_k, it fits a polynomial of degree d via least squares.

The edge response is the maximum absolute normal derivative over all orientations:

$$R_{WVF}(\mathbf{p}) = \max_{k=1,\ldots,N_s} \left|\hat{c}_2^{(k)}\right|$$

Three tunable parameters:

- N_p: support size (number of neighbor pixels)
- N_s: number of candidate orientations
- d: polynomial degree

LINE FILTER (LF)

The LF extends WVF by chaining $2m+1$ WVF evaluations along a line at each orientation. Virtual expansion points are placed at:

$$\mathbf{v}_j^{(k)} = \mathbf{p} + j(\cos\theta_k, \sin\theta_k), \quad j = -m, \ldots, m$$

Derivative estimates are combined via Gaussian-weighted averaging:

$$R_{LF}^{(k)}(\mathbf{p}) = \left| \sum_{j=-m}^{m} w_j \hat{c}_{2,j}^{(k)} \right|$$

Additional parameter m (line half-width) controls tangential averaging. At $m = 14$ (published), LF is $\sim$29$\times$ slower than WVF.

EXPERIMENTAL SETUP

Table: Datasets used in the evaluation (330 images total).

Dataset	Images	Resolution
UDED	30	variable
BIPED v1	50	1280×720
BIPED v2	50	1280×720
BSDS500	200	481×321

- WVF grid: 1,206 valid configurations ($N_p \times N_s \times d$)
- LF grid: 168 configurations (adding m)
- Classical baselines: 54 configurations across 7 operator families
- DL baselines: DexiNed, TEED, PiDiNet, NBED, DiffusionEdge

PARAMETER ABLATION

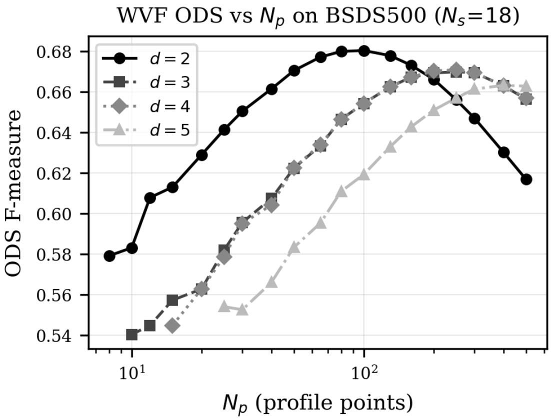
Published parameters are substantially suboptimal

SUPPORT SIZE DOMINATES

Key findings from single-image ablation:

1. ODS peaks at $N_p = 30$–50, not 250
2. N_s saturates by 4–6 orientations
3. $d = 2$ consistently outperforms $d = 4$

At published $N_p = 250$: ODS $\approx$ 0.71 versus optimal 0.86 — a 0.15 gap.

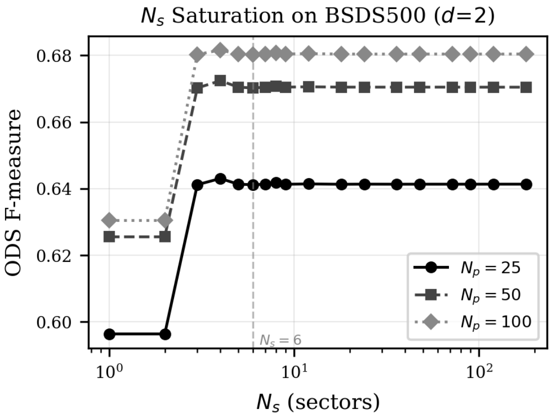


ORIENTATION SATURATION

N_s has minimal effect beyond 4–6 orientations.

The difference between $N_s = 6$ and $N_s = 120$ is typically less than 0.005 ODS.

Reducing N_s from 18 to 4 yields a $4.5\times$ speedup with no loss in edge quality.

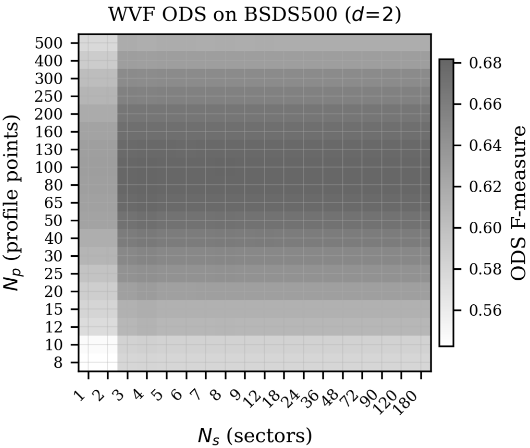


N_p VS N_s HEATMAP

The dominant variation is along N_p.

The optimal region forms a horizontal band at $N_p \approx 25$–50, largely invariant to N_s.

Support size is the critical parameter.



OPTIMIZED VS. PUBLISHED

Table: ODS improvement from parameter optimization across all datasets.

Dataset	Best WVF	Bagan WVF	Δ	Best LF	Bagan LF	Δ
UDED	0.899	0.847	+0.052	0.887	0.729	+0.158
BIPED v1	0.812	0.767	+0.045	0.802	0.643	+0.159
BIPED v2	0.830	0.783	+0.047	0.819	0.658	+0.161
BSDS500	0.682	0.671	+0.011	0.671	0.615	+0.056

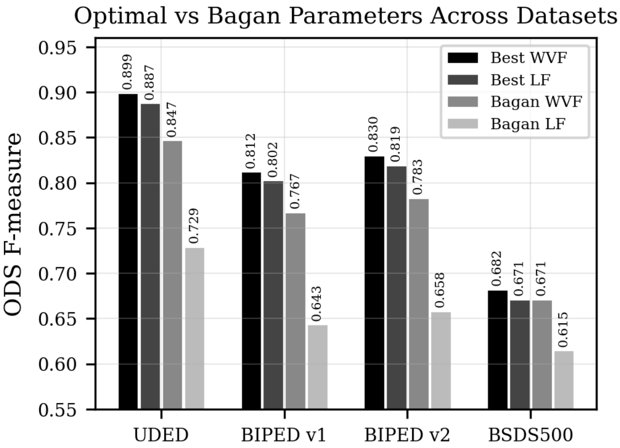
Best WVF: $(\mathrm{N_p} = 25, \mathrm{N_s} = 4, \mathrm{d} = 2)$ on three datasets. Computational savings: $112\times$ reduction over published settings.

CROSS-DATASET COMPARISON

The optimized parameters consistently outperform Bagan's published settings on every dataset.

The WVF gains range from +0.011 (BSDS500) to +0.052 (UDED).

The LF gains are 3–15$\times$ larger, ranging from +0.056 to +0.161, reflecting the greater suboptimality of m $= 14$.



OPTIMAL CONFIGURATIONS

Table: Best-found parameters for each dataset and filter type.

Dataset	Filter	N_p	N_s	d	ODS
UDED	WVF	25	4	2	0.899
UDED	LF	25	36	4	0.887
BIPED v1	WVF	25	4	2	0.812
BIPED v1	LF	100	18	4	0.802
BIPED v2	WVF	25	4	2	0.830
BIPED v2	LF	100	18	4	0.819
BSDS500	WVF	100	4	2	0.682
BSDS500	LF	250	18	4	0.671

WVF consistently selects $d = 2$ and $N_s = 4$. BSDS500 prefers larger N_p (100 vs. 25), likely due to lower image resolution.

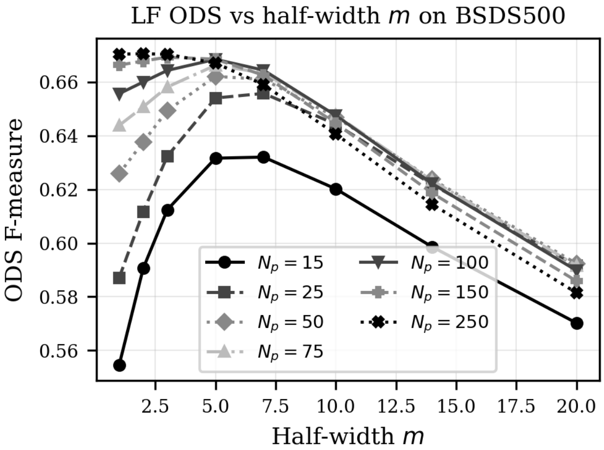
LF HALF-WIDTH m

On all datasets, the best ODS occurs at m $= 1$–3.

The published m $= 14$ yields the worst performance.

On clean data, line-averaging harms edge localization rather than helping it.

WVF outperforms LF by 0.010–0.012 ODS on every dataset.

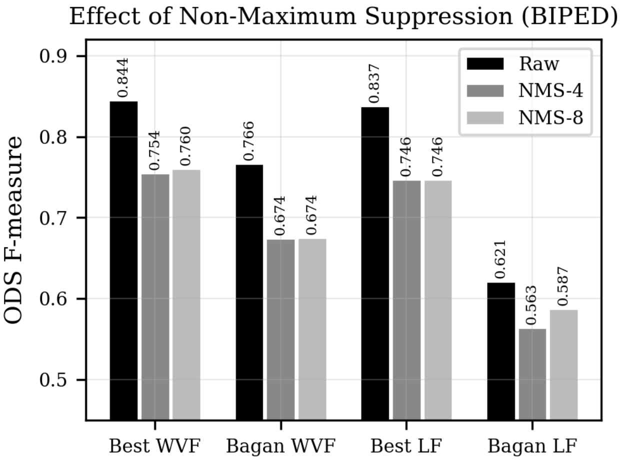


NMS HURTS PERFORMANCE

NMS reduces ODS in all cases.

Best WVF without NMS: 0.844. With 4-dir NMS: 0.754. With 8-dir NMS: 0.760.

Raw gradient maps contain soft, multi-pixel responses that match ground truth within r $= 3$ pixel tolerance. NMS thins these and shifts positions away from annotations.



BENCHMARKING

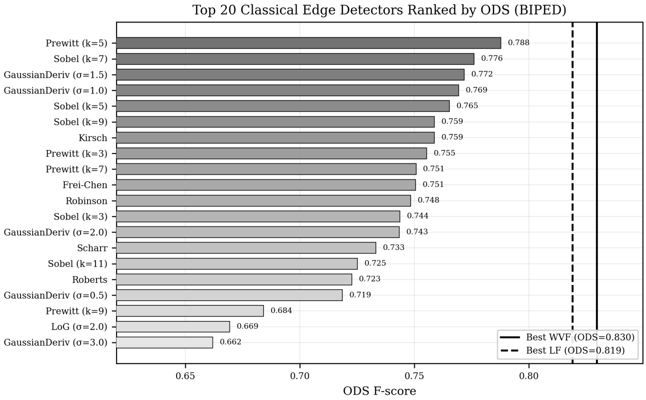
WVF vs. classical operators and deep learning

WVF VS. CLASSICAL OPERATORS

On BIPED v1:

- Best WVF: 0.812
- Best classical (Prewitt-5): 0.788
- Gap: +0.024 ODS (3.0%)

WVF exceeds every classical configuration tested, spanning Sobel, Prewitt, LoG, DoG, Kirsch, Frei-Chen, and Scharr.

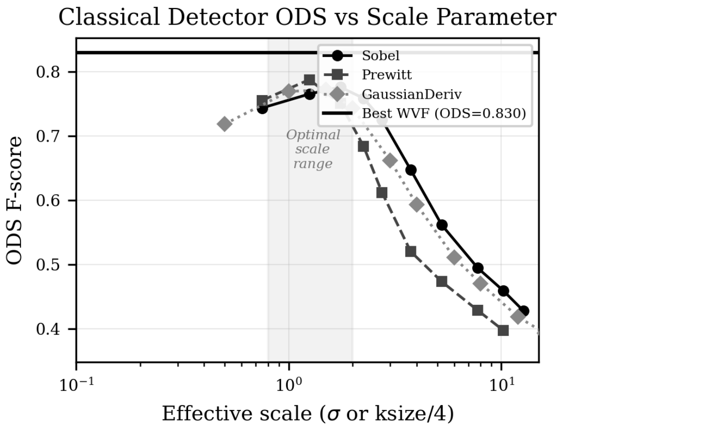


CLASSICAL SCALE SENSITIVITY

Every classical operator family shows a clear optimal scale — performance degrades sharply beyond it.

Sobel-51 scores only 0.428 versus Sobel-7 at 0.776.

WVF's polynomial fitting performs a form of adaptive scale selection, explaining its advantage.



CLASSICAL TOP 15

Method	Parameters	ODS
WVF (optimized)	$\mathsf{N_p} = 25, \mathsf{N_s} = 4, \mathsf{d} = 2$	0.812
LF (optimized)	$\mathsf{N_p} = 100, \mathsf{N_s} = 18, \mathsf{m} = 1$	0.802
Prewitt	$\mathsf{k} = 5$	0.788
Sobel	$\mathsf{k} = 7$	0.776
GaussianDeriv	$\sigma = 1.5$	0.772
GaussianDeriv	$\sigma = 1.0$	0.769
Sobel	$\mathsf{k} = 5$	0.765
Kirsch	–	0.759
Prewitt	$\mathsf{k} = 3$	0.755
Frei-Chen	–	0.751
Robinson	–	0.748

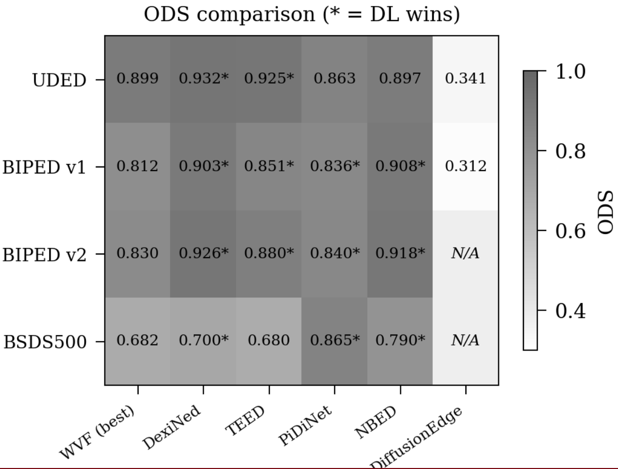
DL ON CLEAN DATA

On clean data, DL models generally outperform WVF.

Per-image ODS on BSDS500 (5-image):

- PiDiNet: 0.876
- DiffusionEdge: 0.795
- NBED: 0.795
- DexiNed: 0.737
- TEED: 0.669

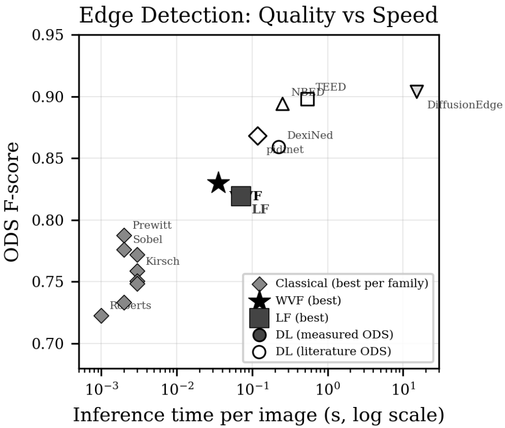
Per-image WVF (oracle): 0.907. Full-dataset WVF: 0.682.



COMPUTATIONAL COST

Method	Time	Rel.
WVF	0.036s	$1.0\times$
PiDiNet	0.119s	$3.3\times$
DexiNed	0.226s	$6.3\times$
NBED	0.253s	$7.1\times$
TEED	0.538s	$15\times$
DiffusionEdge	15.22s	$427\times$

WVF: real-time at 30+ fps.



NOISE ROBUSTNESS

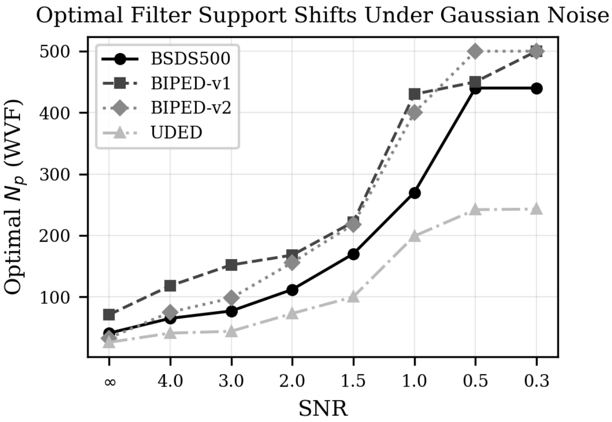
WVF adapts; deep learning collapses

N_p SHIFTS UNDER NOISE

Under severe Gaussian noise (SNR = 0.3):

- Optimal N_p shifts from $\sim$40 to $\sim$440
- A $\sim 10\times$ increase in support size
- Partially vindicates Bagan's $N_p = 250$

However, $d = 2$ remains universally optimal — $d = 4$ is suboptimal at every SNR level.

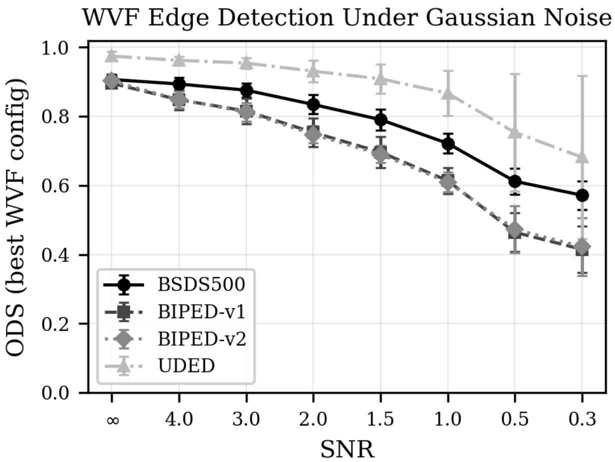


WVF NOISE DEGRADATION

Degradation profiles vary by noise type:

- Speckle: least destructive (23% drop)
- Poisson: intermediate (30%)
- Salt-and-pepper: intermediate (31%)
- Gaussian/uniform: most severe (36–37%)

Ranking is consistent across all datasets.



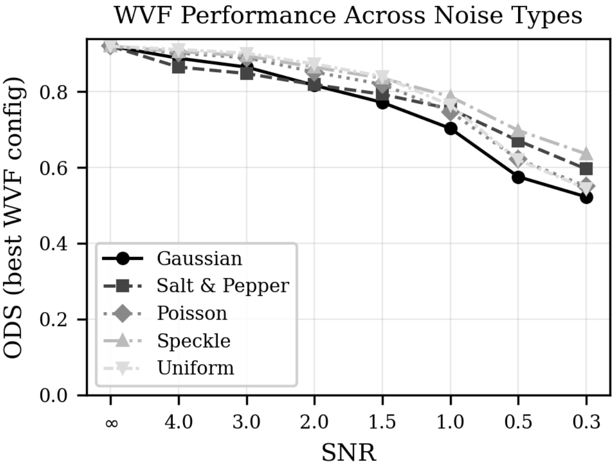
NOISE CONSISTENCY

The relative ordering of noise types is consistent across all four datasets.

Ranking (least to most destructive):

Speckle $>$ Poisson $\approx$ Salt-and-pepper $>$ Uniform $\approx$ Gaussian

The parameter shift magnitude is dataset-dependent: UDED (high-contrast underwater) requires a more modest N_p increase than BIPED (fine-textured edges).

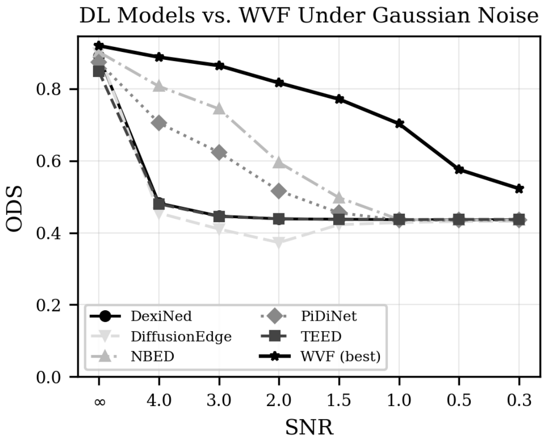


DL COLLAPSE UNDER NOISE

All 5 DL models degrade severely:

- PiDiNet: 0.876 → 0.329 (62% loss)
- DexiNed: near-total collapse
- TEED: severe degradation
- NBED: most graceful decline
- DiffusionEdge: erratic behavior

None include noise augmentation in training.

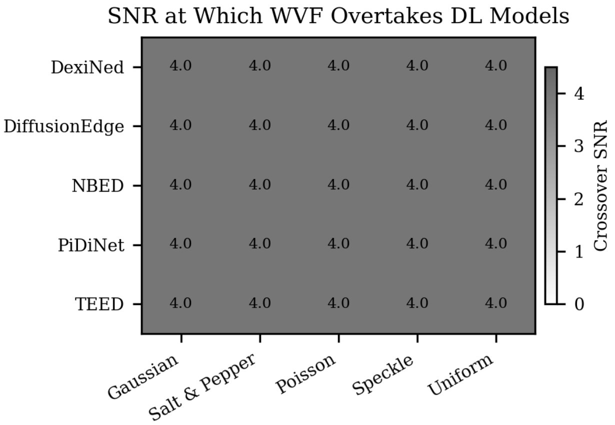


WVF OVERTAKES DL

Crossover analysis:

- DexiNed, TEED: overtaken at all SNR levels
- PiDiNet: overtaken at SNR $\approx$ 4.0
- NBED: most competitive, still exceeded

At SNR = 0.3: WVF beats every DL model on every test image (100% crossover).

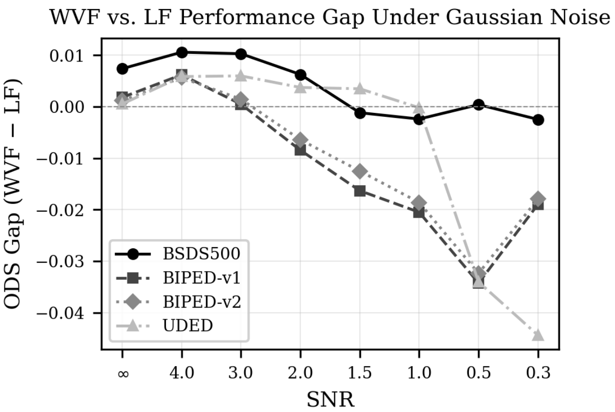


WVF VS. LF UNDER NOISE

On clean data, WVF leads by $\sim$0.007 ODS.

Under severe noise (SNR $\leq$ 1.0), the gap closes and LF slightly overtakes WVF for Gaussian, salt-and-pepper, and uniform noise.

The reversal is modest (0.002–0.009 ODS) — the LF's line-averaging provides marginal noise resilience.



STATISTICAL VALIDATION

Nonparametric tests confirm all major claims

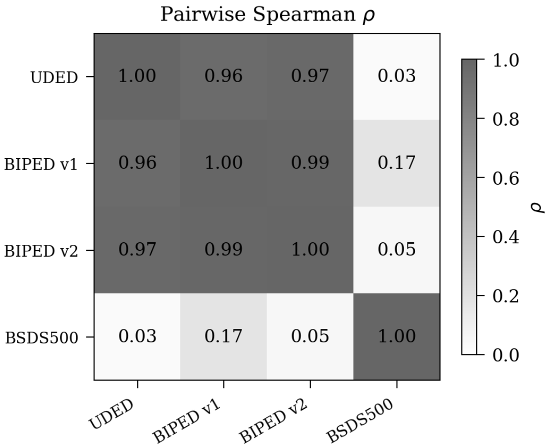
RANKING CONSISTENCY

Kendall's W $= 0.647$ (moderate concordance).

UDED, BIPED v1, BIPED v2: $\rho > 0.96$ (near-perfect agreement).

BSDS500 is a strong outlier: $\rho = 0.03$–0.17 with all other datasets.

Top-10 configurations on BSDS500 share zero overlap with other datasets.



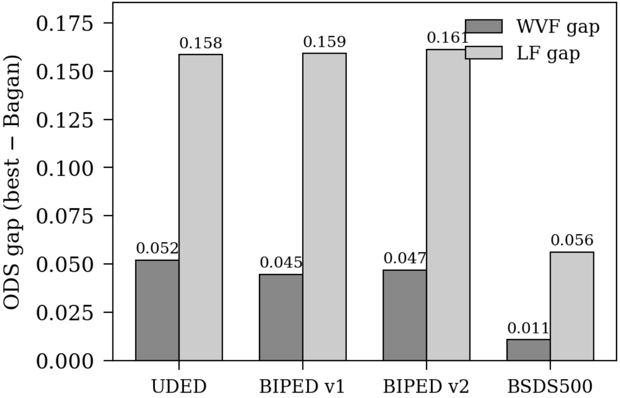
ODS GAPS

All eight gaps (4 datasets $\times$ 2 filters) are positive.

Sign test: p $= 0.0625$ for both filters (minimum achievable with n $= 4$).

The LF gaps are 3–15$\times$ larger than WVF gaps, reflecting m $= 14$ as the primary source of suboptimality.

Even the smallest gap (+0.011 on BSDS500) is practically meaningful across 200 images.



WVF VS. LF SIGNIFICANCE

Table: Wilcoxon signed-rank test: WVF vs. LF at matched parameters.

Dataset	Median diff	Cohen's d	T	p
UDED	+0.0003	-0.46	45	0.805
BIPED v1	-0.0009	-0.77	21	0.989
BIPED v2	-0.0003	-0.64	33	0.940
BSDS500	-0.0034	-1.15	6	1.000

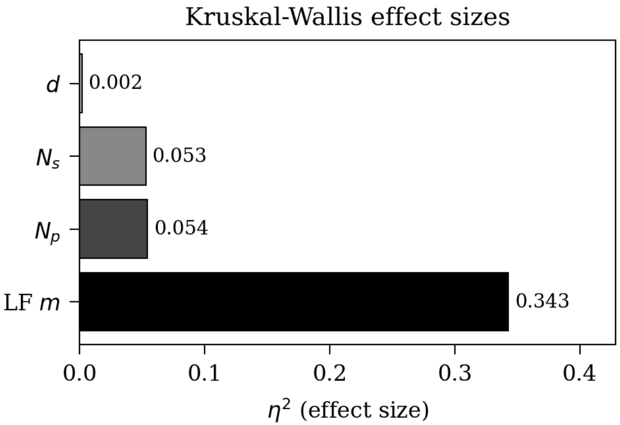
No significant difference on any dataset (p > 0.8 everywhere). The median ODS difference is less than 0.004 on every dataset. At matched parameters, WVF and LF produce essentially identical edge maps.

PARAMETER SENSITIVITY

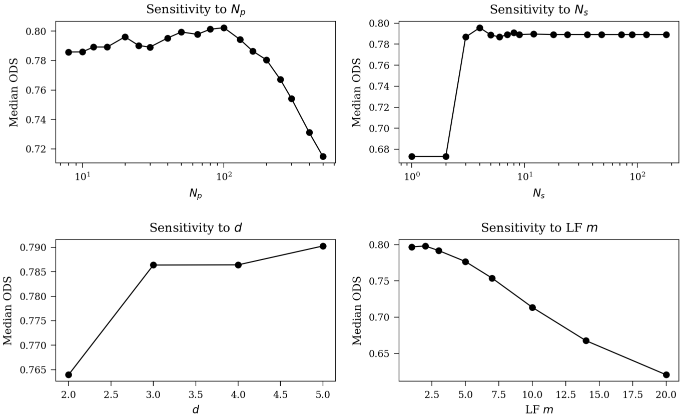
Kruskal–Wallis analysis reveals parameter priority:

Parameter	η^2
m (LF half-width)	0.343
N_p (support size)	$\sim$0.05
N_s (orientations)	$\sim$0.05
d (poly. degree)	$<$0.01

m is by far the most influential — the LF's line extension is its biggest liability on clean data.



MEDIAN ODS BY PARAMETER



N_p: inverted-U peaking near 80–100. N_s: saturates at 3–4. d: minimal effect. LF m: monotonic decline beyond $m = 2$.

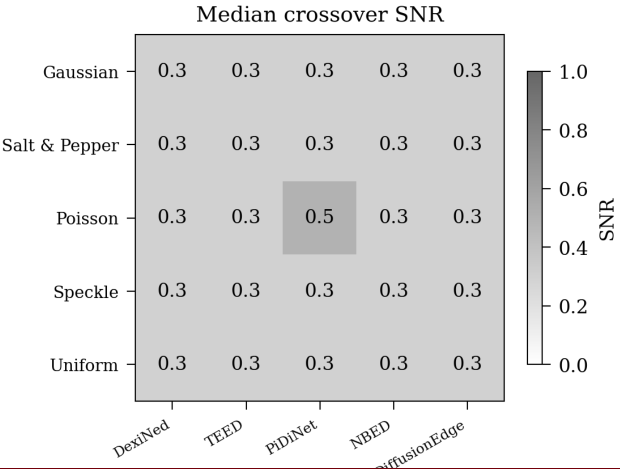
CROSSOVER HEATMAP

Median crossover SNR for each noise type $\times$ DL model combination.

23 of 25 combinations show crossover at SNR = 0.3 (the most extreme level tested).

100% of image pairs show WVF overtaking DL at SNR $\leq$ 0.5.

Bootstrap 95% CIs are narrow: $[0.30, 0.30]$ for most combinations.



EVIDENCE SUMMARY

Table: Evidence strength for each major claim.

Claim	Strength	Key Statistic
Optimized $>$ Bagan	Strong practical	p $= 0.0625$, gaps 0.01–0.16
WVF $=$ LF (matched)	Strong null	p > 0.8, $\|\Delta\| < 0.004$
m most influential	Very strong	$\eta^2 = 0.343$
N_p, N_s moderate	Strong	$\eta^2 \approx 0.05$
BSDS500 diverges	Very strong	$\rho = 0.03$–0.17
DL $>$ WVF (clean)	Consistent	3–4 of 4 datasets
WVF $>$ DL (noise)	Very strong	100% crossover at SNR ≤ 0.5

DISCUSSION

Reconciling findings and looking forward

RECONCILING BAGAN

Bagan and Wang's parameter choices are simultaneously suboptimal and well-motivated.

On clean data

- N_p: 250 $\rightarrow$ 25–100 (substantial gain)
- d: 4 $\rightarrow$ 2 (universal improvement)
- N_s: 18 $\rightarrow$ 4 (no loss, $4.5\times$ speedup)
- m: 14 $\rightarrow$ 1–3 (most damaging choice)

Under noise

- $N_p = 250$ is near-optimal at SNR = 0.3 under speckle noise
- Physical mechanism: larger neighborhoods average more noise samples
- Bias–variance trade-off shifts in favor of larger support under noise

The one universally suboptimal choice: $d = 4$. Higher-order polynomials overfit noise fluctuations at every SNR level.

BSDS500 OUTLIER

The near-zero Spearman correlation between BSDS500 and other datasets is consequential.

- Root cause: small image size (481×321 vs. 1280×720), diverse content, multi-annotator ground truth with varying criteria
- WVF's circular neighborhood at $\mathrm{N_p} = 100$ spans a proportionally larger fraction of BSDS500 images
- Top-10 configurations share zero overlap across BSDS500 and other datasets

Implication: Benchmark results on BSDS500, which dominate the edge detection literature, should not be extrapolated to other imaging domains without validation. Practitioners should tune on domain-representative data.

WVF'S NICHE

Against classical operators

- Unambiguous: outperforms all 54 configs
- 2.4 ODS points over best classical (Prewitt-5)
- Polynomial fitting provides adaptive scale selection

Against deep learning

- Condition-dependent comparison
- DL wins on clean data (learned semantics)
- WVF wins under noise (parametric adaptability)
- $3.3\times$–$427\times$ faster, training-free, fully interpretable

WVF occupies a practical niche: highest accuracy of any training-free method, real-time capable, and robust to degraded conditions.

LIMITATIONS

- Noise ablation uses only 5 images per dataset (qualitative trends robust, specific crossover SNR values have uncertainty)
- WVF crossover uses per-image oracle parameter selection, overestimating deployable performance
- DL models evaluated with published weights only (no noise-augmented retraining)
- Classical baseline comparison covers only BIPED v1
- All noise models are synthetic — real underwater noise has complex spatial correlations
- $n = 4$ dataset design limits statistical power (minimum sign-test $p = 0.0625$)
- LF ablation fixed $d = 4$; exploring $d = 2$ for LF is important future work

CONCLUSION

KEY FINDINGS

1. Published parameters are suboptimal: $N_p = 25–100$, d $= 2$, $N_s = 4$ improves ODS by 0.01–0.16 with $112\times$ computational savings
2. WVF beats all 54 classical operators by +2.4 ODS points
3. WVF is $3.3\times–427\times$ faster than every DL model tested
4. Under noise, optimal N_p shifts to 250–500, partially vindicating Bagan and Wang's design
5. WVF overtakes DL models at mild noise (SNR $\approx$ 4.0); 100% crossover by SNR = 0.3
6. BSDS500 is a statistical outlier: its rankings are uninformative about other domains ($\rho = 0.03–0.17$)

RECOMMENDATIONS

Clean imagery

- Use WVF (not LF)
- $N_p = 25$–50 (high-res) or 50–100 (low-res)
- $d = 2$, $N_s = 4$
- Skip NMS (raw gradient maps score higher)

Noisy environments

- Increase N_p proportionally to noise level
- Moderate noise: $N_p = 65$–112
- Severe noise: $N_p = 250$–500
- $d = 2$ remains optimal at all SNR levels

Edge detection method selection is fundamentally condition-dependent. WVF's principal strength is parametric adaptability across the full spectrum of operating conditions.

WVF VS. DL: DECISION GUIDE

Table: Decision guide: WVF vs. deep learning for edge detection.

Criterion	WVF	DL Models
Clean-data accuracy	Good	Best
Noisy-data accuracy	Best	Poor
Computational speed	Real-time	3–$427\times$ slower
Training data required	None	Large annotated sets
Interpretability	Full	Opaque
Parameter adaptability	Runtime	Requires retraining

For safety-critical applications (medical, autonomous navigation, industrial inspection), WVF's complete interpretability is a significant advantage.

FUTURE DIRECTIONS

1. Full-dataset noise ablation: extend to all 330 images across all noise conditions ($\sim$2,500 GPU-hours)
2. Adaptive WVF: automatic N_p adjustment based on estimated noise level via lookup table
3. LF with $d = 2$: WVF results strongly suggest this would improve LF performance
4. Noise-augmented DL retraining: fairer comparison under noise conditions
5. Real-world validation: field testing on actual underwater, medical, and industrial imagery
6. Additional benchmarks: a fifth dataset would lower minimum sign-test p to 0.03125

THANK YOU!

Questions & Discussion

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